

Virtual Reality-based Privacy Risk Assessment and Threat Mitigation in Federated Learning for Facial Recognition



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1. Introduction

VR technology allows for immersive user interactions through extensive data collection, enhancing user experience but raising significant privacy concerns:

- VR systems collect detailed data such as body motion, eye gaze, hand movements, and facial expressions.
- This data enables user identification and behavior analysis, making VR susceptible to privacy attacks.
- Facial recognition and Facial Expression Recognition (FER) is particularly vulnerable to attacks due to sensitivity of biometric data.

Examples of Privacy Attacks in VR:

- **Poisoning Attacks:** Involves injecting malicious data to corrupt models, allowing adversaries to impersonate users or gain unauthorized access.
- **De-anonymization Attacks:** Using biometric and behavioral data to re-identify users despite anonymity through avatars
- **Man-in-the-Room (MitR) Attacks:** Allows adversaries to invisibly connect to private VR rooms to violate privacy of users and execute malware, like worms.

3. Overview of Security Attacks on a VR System Using FL

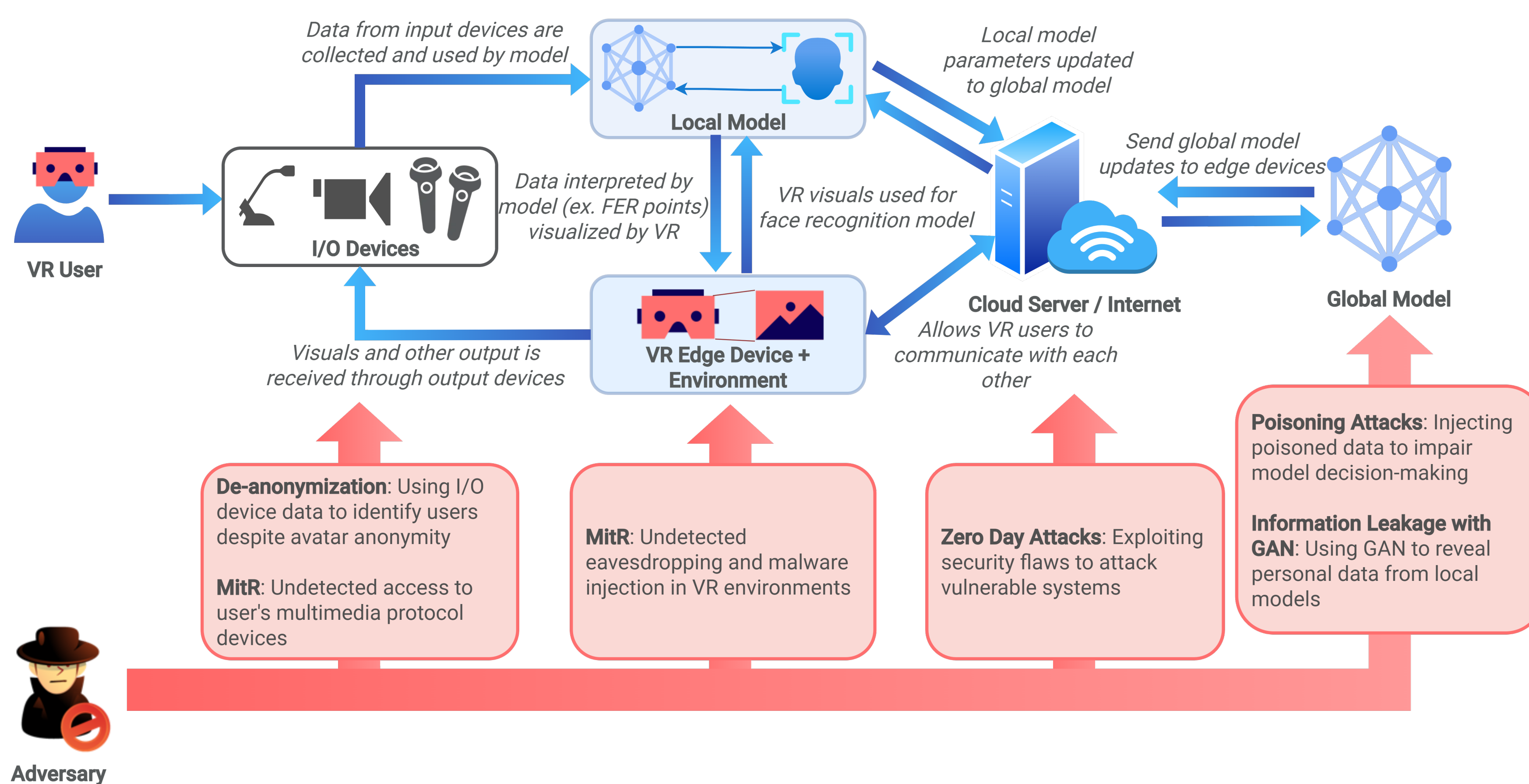


Figure 2: This flowchart demonstrates how different attacks can occur at different points of a VR's system that utilizes an artificial intelligence model using federated learning as well as how it may affect these parts of the system.

5. Methodology

Step 1. We implement a face recognition program that works on static images. It is important that the model works well on personal images, which may not be as abundant as celebrities, for training. Therefore, we are utilizing Adam Geitgey's Python package, face-recognition, which allows for one-shot learning.

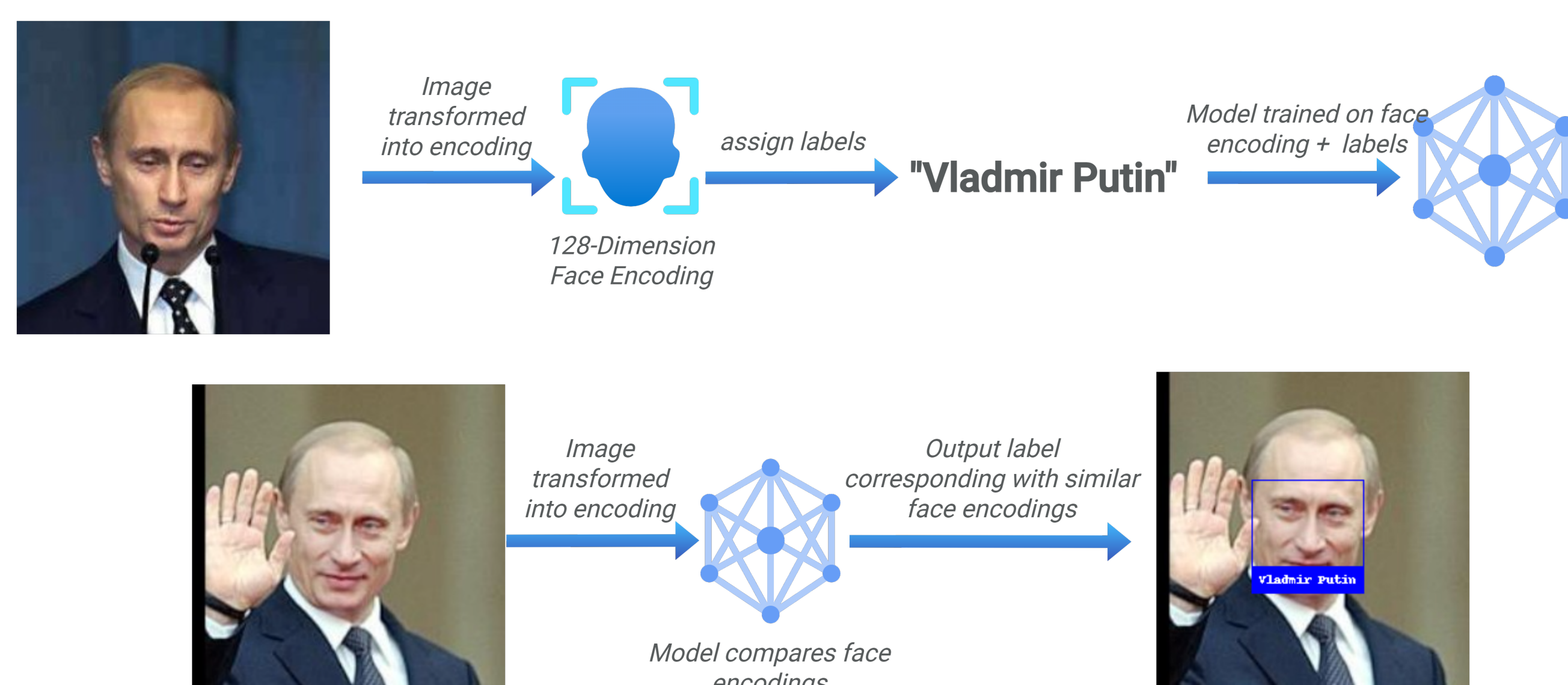


Figure 4: How the model utilizes face encodings from an image to train itself.

Step 2. Building up, we transformed this model to work using a live webcam. To maximize efficiency, we process every other frame for facial recognition analysis. This step demonstrates that using live webcam with facial recognition can eventually be used in a VR setting, whether it is using the headset's camera or projecting the headset's view onto a computer and using this program to recognize faces from the VR environment.

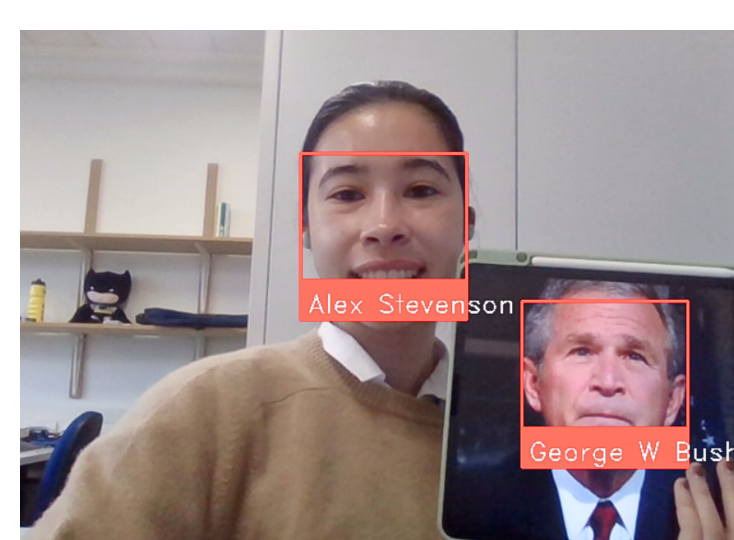


Figure 5: Screenshot from live facial recognition.

2. Federated Learning

Federated Learning (FL) can be a countermeasure in VR systems:

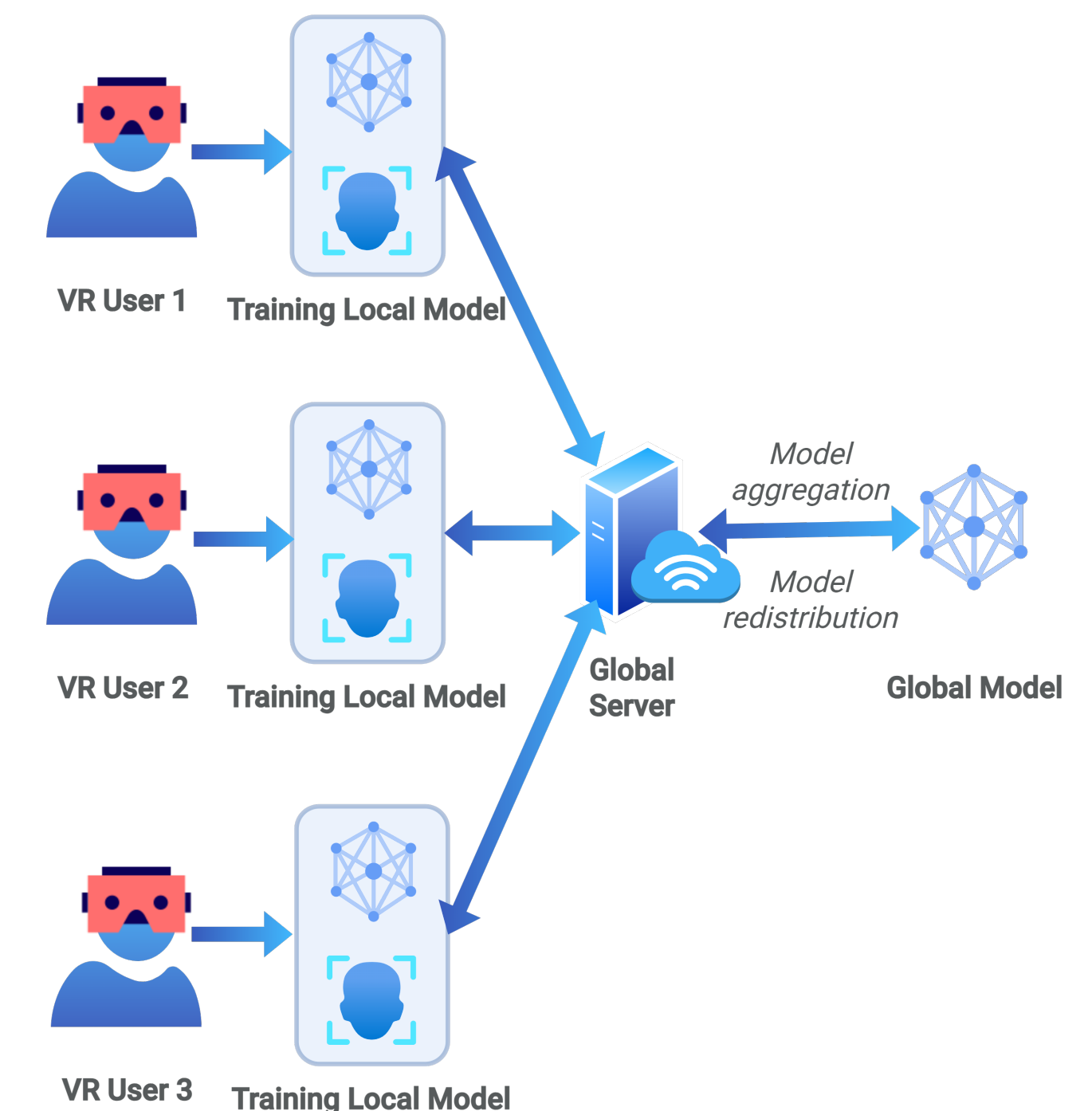


Figure 1: FL model used with VR users.

FL must be thoroughly evaluated as it may not suffice for more aggressive privacy attacks.

4. Poisoning Attacks Example

Figure 3 shows how poisoning attacks can affect a model in FL and how it can affect victims of this attack. This project eventually aims to use FL with facial recognition to see how privacy attacks like poisoning attacks can affect the model in a VR setting.

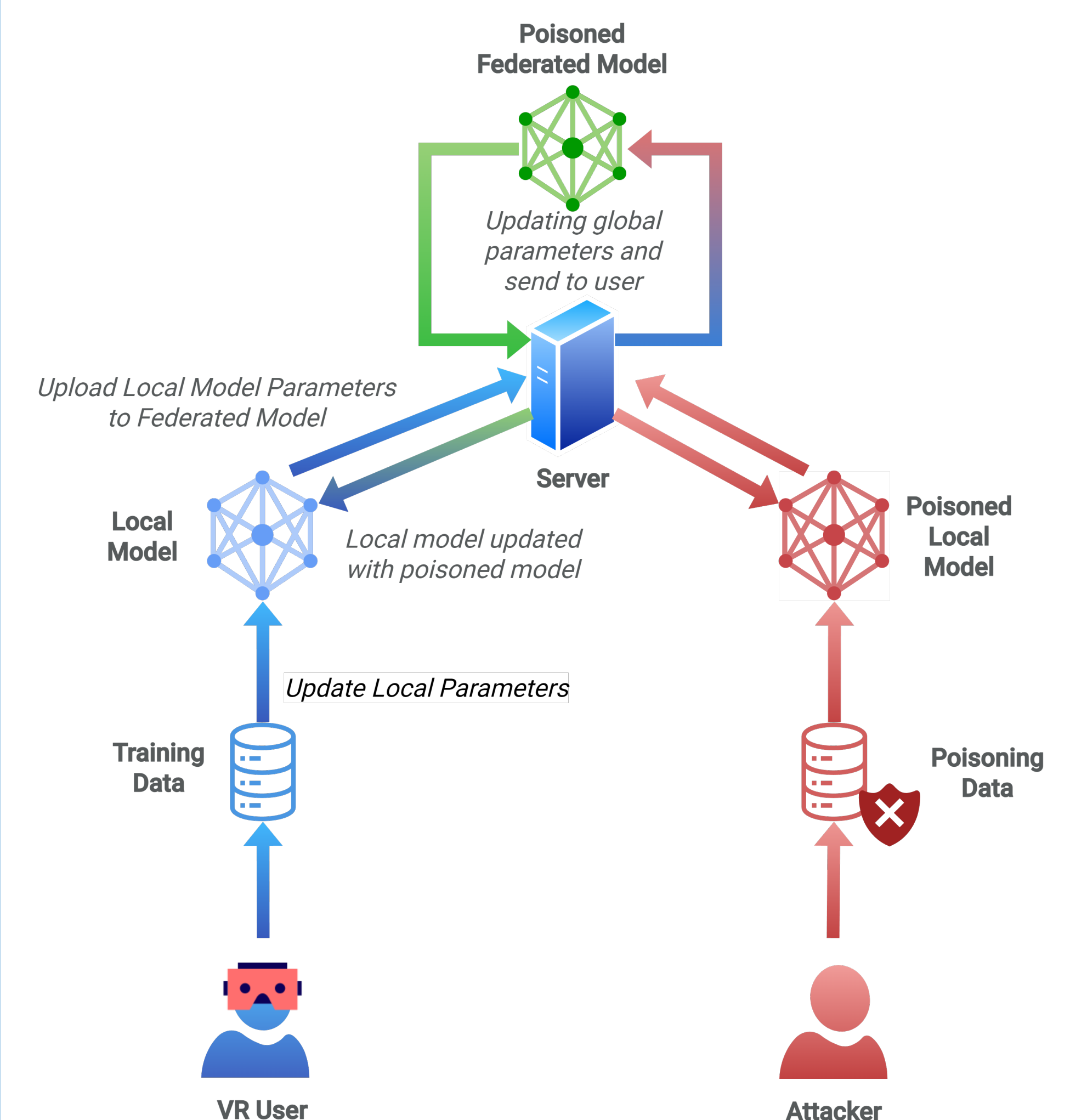


Figure 3: Poisoning attacks on a VR system using federated learning.

6. Future Work

- Store updated model with facial encodings and labels on the VR device.
- Implement Python live facial recognition program using the VR webcam.
- Experiment with different attacks to assess FL vulnerability in VR.

Related Publications

- [1] Chen, Y., Huang, S., et al. Federated learning for metaverse: A survey. In Companion Proceedings of the ACM Web Conference 2023.
- [2] Jinghuai Lin and Marc Erich Latoschik. 2022. Digital body, identity and privacy in social virtual reality: A systematic review. Front. Virtual Real. 3, (November 2022).